

RESEARCH INTEREST

Machine Learning

- *Data-Centric AI; Reinforcement Learning; Extrapolation; Robustness; Causality; AI for Social Good*

TECHNICAL SKILLS

Skills

- *Python; PyTorch; Hugging Face Transformers; SQL; Git; Linux; data pipelines; LLM synthetic data; benchmarking/evaluation; NLP*

EDUCATION

Swiss Federal Institute of Technology (EPFL)

Lausanne, Switzerland

- *M.Sc. in Computer Science (Specialization in Computer Science Theory)* *Sept. 2024 – Present*
 - **Coursework:** Machine Learning; Advanced Algorithms; Causal Thinking; Mathematics of Data: From Theory to Computation; Markov Chains and Algorithmic Applications; Deep Learning; Reinforcement Learning; Modern Natural Language Processing; Distributed Algorithms; Digital Education
 - Completed mandatory military service (Mar. 2025 – Jun. 2025)

National Taiwan University

Taipei, Taiwan

- *M.Sc. in Data Science* *Feb. 2023 – June 2024*
 - Paused studies after June 2024 to pursue a master's degree at EPFL
 - **Coursework:** Machine Learning; Natural Language Processing; Statistical Foundations of Data Science (I); Computational Methods for Data Science; Big Data Systems

National Taiwan University of Science and Technology

Taipei, Taiwan

- *B.Sc. in Applied Science and Technology, Major in Computer Science* *Sept. 2018 – June 2022*
 - Graduated 1st in the program (2022)
 - **Coursework:** Statistics (I); Probability and Statistics; Discrete Mathematics; Linear Algebra; Calculus (I); Algorithms; Data Structures; Database Systems; Operating Systems; Digital Logic Design; Computer Networks; Webpage Development; Object-Oriented Programming; Introduction to Computers; Value of AI and Data; Introduction to Fuzzy Systems; Machine Learning Foundations

RESEARCH AND INDUSTRY EXPERIENCE

DLab, EPFL

Lausanne, Switzerland

- *Semester Project Student* *Sept. 2026 – Present*
 - **Advisors:** Robert West; Raghav Singhal
 - Investigating pre-training signals for predicting post-training performance, ranking, and generalization across models and data regimes.

MLO Lab, EPFL

Lausanne, Switzerland

- *Semester Project Student* *Sept. 2025 – Present*
 - **Advisors:** Martin Jaggi; El Mahdi Chayti
 - Studying looped/recurrent Transformers for length extrapolation, focusing on learned stochastic stopping, OOD robustness, and stability across inference-time computation depths.

Robust Machine Learning Group, Max Planck Institute for Intelligent Systems

Tübingen, Germany

- *Research Intern* *June 2025 – Aug. 2025*
 - **Advisors:** Wieland Brendel; Patrik Reizinger
 - Designed controlled experiments on OOD extrapolation in looped Transformers, analyzing the effects of initialization, data order, and loop budget on generalization stability.

CKIP Lab, Academia Sinica & IR Lab, National Taiwan University

Taipei, Taiwan

- *Research Assistant* *Feb. 2023 – June 2024*
 - **Advisors:** Wei-Yun Ma; Pu-Jen Cheng
 - Developed data-weighting methods for LLM-generated synthetic text-classification data, improving robustness under source-target distribution shift.

Industrial Technology Research Institute

▪ Intern

- Engineered short- and long-term load forecasting methods for Taiwan Power Company.

Wireless System Lab, National Taiwan University of Science and Technology

▪ Research Assistant

- **Advisor:** Chin-Ya Huang
- Devised an MTC scheme for dynamic CBRS channel allocation.
- Implemented RLNC and Galois field arithmetic in P4; simulated with ONOS/Mininet.

Taipei, Taiwan

July 2021 – Sept. 2021

Taipei, Taiwan

Sept. 2019 – Jan. 2021

PAPERS

- [P1] **Hsun-Yu Kuo**, El Mahdi Chayti, Patrik Reizinger, Wieland Brendel & Martin Jaggi, “Stabilizing Extrapolation in Looped Transformers via Learned Stochastic Stopping”. In *The Fortieth Annual Conference on Neural Information Processing Systems* (NeurIPS 2026) [📄](#)
- [P2] **Hsun-Yu Kuo**, Yin-Hsiang Liao, Yu-Chieh Chao, Wei-Yun Ma & Pu-Jen Cheng, “Not All LLM-Generated Data Are Equal: Rethinking Data Weighting in Text Classification”. In *Proceedings of the 13th International Conference on Learning Representations* (ICLR 2025) (**Spotlight**, top 5.1%) [📄](#)
- [P3] **Hsun-Yu Kuo**, Szu-Yu Liu, Chin-Ya Huang, Yu-Chi Chen & Meng-Hua Xie, “Reliable Data Transmission through Private CBRS Networks”. In *Preprint* (2023) [📄](#)
- [P4] **Hsun-Yu Kuo**, T.-W. Liu, Y.-P. Huang, *et al.*, “Differential Diagnostic Value of Machine Learning–Based Models for Embolic Stroke”. *Clinical and Applied Thrombosis/Hemostasis* (2023)
- [P5] Hsuan-Min Wang, Yo-Ping Huang, **Hsun-Yu Kuo**, *et al.*, “The modified spatial context memory test for assessing cognitive aging in middle-aged and older adults”. *Neuroscience Research* (2019)

TEACHING EXPERIENCE

Teaching Assistant, National Taiwan University

▪ System Programming

- Led TA sessions and supported 100+ students.
- Designed a context-switch simulation using non-local jumps and signals for a course assignment.

Taipei, Taiwan

Sept. 2023 – Jan. 2024

HONORS AND AWARDS

Phi Tau Phi Scholastic Honor Society of the Republic of China

▪ Honorary Membership

Taiwan

2022

Mengjia Longshan Temple Scholarship

▪ Academic Excellence Scholarship (Scholarship for Excellent Academic Performance)

Taiwan

2021

Academic Excellence Award

▪ Seven consecutive awards for excellent academic performance

Taiwan

2018–2022

Kanagawa International Science Forum

▪ 1st Place (Outstanding Poster Presentation Award) – Engineering Category

Kanagawa, Japan

2017

National Skills Competition

▪ 1st Place (Golden Award) – Software Applications Development

Taiwan

2016

PROFESSIONAL SERVICE

Conference Reviewer

▪ ICLR 2026, NeurIPS 2026, ICLR 2027

Volunteer

▪ SIGIR 2023